

dd/mm is your birthday. Let $d=dd$ (example $dd=03$, $d=3$) and $m=mm$.

Question 1. (20 points) The weights of certain machine components are normally distributed with a mean of $(100+d)$ gram and a standard deviation of m gram. Find the d percentile (the value separate the top $(100-d)\%$ and the bottom $d\%$. (*These weights could serve as limits used to identify which components should be rejected.*)

Question 2. (20 points) Given $X=N(\mu;\sigma^2)$ where $\mu=(d+m)$ and $\sigma=m$. Find the following probabilities:

- a) $P(X>d+2m)$
- b) $P(X<d-m)$

Question 3. (30 points) An Associates Market Research found that $(d+m)\%$ of Americans do not think that having a college education is important to succeed in the business world. If a random sample of fifty Americans is selected, find these probabilities.

- a. Exactly m people will agree with that statement.
- b. At most $m+1$ people will agree with that statement.
- c. Fewer than d people will agree with that statement.

Question 4. (30 points) You are invited to participate in a game at a carnival. Here are the rules:

Entry Fee: You pay \$10 to play.

Game Mechanics: The game involves drawing a card randomly from a standard deck of 52 playing cards and then spinning a fair wheel divided into 4 equally likely outcomes (A, B, C, D).

Rewards:

If you draw a spade:

- Spin result A: You win \$50.
- Spin result B: You win \$30.
- Spin result C: You win \$10.
- Spin result D: You win nothing.

If you draw a red card (hearts or diamonds):

- Spin result A: You win \$20.
- Spin result B: You win \$15.
- Spin result C: You win \$5.
- Spin result D: You win nothing.

If you draw a club:

- Spin result A: You win \$40.
- Spin result B: You win \$20.
- Spin result C: You win \$10.
- Spin result D: You win nothing.

Profit: Your profit is your total winnings minus the \$10 entry fee.

- a) Calculate the expected profit of this game based on the probability distribution of outcomes.

Is this game fair for the player? Justify your answer.

- b) Suppose the entry fee is reduced to \$8. How would this impact the expected profit? Would the game now be favorable to the player?